

Entanglement generation in a two-body Schrödinger–Newton model

Marcin Płodzień^{1,2}, Julia Oseka-Lenart^{3,4}, Maciej Lewenstein², and Michał Eckstein⁵

¹Qilimanjaro Quantum Tech, Carrer de Vençuela 74, 08019 Barcelona, Spain

²ICFO-Institut de Ciències Fòniques, The Barcelona Institute of Science and Technology,
08860 Castelldefels (Barcelona), Spain

³Astronomical Observatory, Faculty of Physics, Astronomy and Applied Computer Science
Jagiellonian University, Orla 171, Kraków, 30-244, Poland

⁴Doctoral School of Exact and Natural Sciences, Jagiellonian University, ul. Łojasiewicza 11,
30-348 Kraków, Poland

⁵Institute of Theoretical Physics, Faculty of Physics, Astronomy and Applied Computer
Science, Jagiellonian University, ul. Łojasiewicza 11, 30-348 Kraków, Poland

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Abstract

The Schrödinger–Newton (SN) equation provides a semiclassical framework for the evolution of self-gravitating massive quantum systems. We propose a two-body Schrödinger–Newton model that separates local nonlinear self-localization from the non-separable Newtonian pair potential. Analytically, we show that the nonlinear self-field preserves the Schmidt spectrum, whereas direct entanglement generation arises from the nonseparable pair potential. Using numerical simulations in a regularized one-dimensional geometry, we find that entanglement generation depends sensitively on the initial spatial configuration and on the mass ratio. Highly localized, self-bound wavepackets experience minimal entanglement growth during scattering. Spatial delocalization and kinetic dispersion broaden the interaction region, amplifying the entangling power of the pair potential and exciting higher-order spatial modes. For dispersive Gaussian initial states, mass asymmetry shatters the lighter particle, producing Wigner negativity and rapid entanglement growth, whereas stationary SN profiles strongly suppress this effect. Stationary SN profiles isolate the bare pair-potential contribution; dispersive Gaussian initial states inflate it.

1 Introduction

The Schrödinger–Newton (SN) equation aims at modelling the self-gravitational evolution of a quantum wavefunction. In the single-particle case, the Newtonian potential is sourced by the particle’s own probability density, and the resulting nonlinear feedback produces gravitational self-localization, whereby a spreading wavepacket is re-focused by its self-generated gravitational well. This mechanism has been studied extensively as a candidate for explaining the quantum-to-classical transition through gravity [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]. The single-particle SN equation can

be derived from different principles: coupled Schrödinger–Poisson equations [13], the non-relativistic limit of Einstein–Dirac or Einstein–Klein–Gordon systems [14], semi-classical gravity [15, 10] or a Hartree approximation to fully quantum gravity [10]. Independently, the SN equation serves as an effective model to describe Cold Dark Matter [16, 17, 18], boson stars evolution [19] or interactions in ultra cold gases [20, 21, 22, 23, 24, 25].

For more than one particle, the prescription for the dynamics of self-gravitating quantum systems is no longer unique. One must specify whether the classical gravitational field is sourced only through one-body marginal densities, through a shared mean field, or through an explicit pair operator depending on $x_1 - x_2$. These choices are dynamically inequivalent. Marginal-density mean fields preserve product states, whereas a nonseparable pair operator can change the Schmidt spectrum of an initially factorized state. Entanglement-witness proposals aim to directly probe this distinction by testing whether the interaction between two masses can increase their Schmidt rank [26, 27, 28, 29, 30, 31].

The canonical N -particle Schrödinger–Newton equation is based on semiclassical mean-field gravity, $G_{\mu\nu} = 8\pi G/c^4 \langle \hat{T}_{\mu\nu} \rangle$, à la Møller–Rosenfeld [1, 10]. In the Newtonian regime it yields separability-preserving mean-field interactions that preclude entanglement [32]. In contrast, the standard nonrelativistic quantum treatment of two gravitating masses contains a linear nonseparable Newtonian pair potential, analogous to the standard Coulomb electromagnetic interaction (see e.g. [33]). Here we propose a two-body Schrödinger–Newton-type equation that supplements nonlinear semiclassical self-localization with a nonseparable pairwise Newtonian interaction. By including both mechanisms, the present model isolates which term is responsible for entanglement and provides a direct comparison against strictly separability-preserving mean-field reductions.

The proposed model is phenomenological rather than a first-principles relativistic theory. However, any extension that combines Diósi–Penrose-type self-localization [1, 2, 10] with Newtonian mutual attraction must specify whether the mutual interaction is a separability-preserving mean field or a nonseparable pair potential. Formally separating these mechanisms allows us to identify which term changes the Schmidt spectrum and which term only reshapes the one-body marginals.

We then solve the equation numerically for two-particle collisions in 1D. The simulations demonstrate that entanglement growth is driven by the nonadditive Newtonian pair potential. For dispersive Gaussian initial states, mass asymmetry shatters the lighter particle, producing Wigner negativity and rapid entanglement growth. In contrast, in the localized-product mass-asymmetry scan, highly localized stationary profiles strongly suppress this disruption and remain weakly entangled across all simulated mass ratios.

The paper is organized as follows. Section 2 introduces our two-body Schrödinger–Newton model, its effective Hamiltonian structure, and the entanglement measure. Section 3 presents the numerical results, introducing each initial state and its simulation in turn. Section 4 contains the conclusions and an outlook into possible developments of realistic physical models.

2 Two-Body Schrödinger–Newton Model

Our starting point is the Schrödinger–Newton equation [1, 2, 10],

$$i\hbar \partial_t \psi(t, \mathbf{x}) = \left[-\frac{\hbar^2}{2m} \nabla^2 - Gm^2 \int \frac{|\psi(t, \mathbf{y})|^2}{|\mathbf{x} - \mathbf{y}|} d^3 \mathbf{y} \right] \psi(t, \mathbf{x}), \quad (1)$$

modelling the time-evolution of the wavefunction of a single self-gravitating quantum particle of mass m . We consider a 1D analogue of Eq. (1) in dimensionless units (cf. [12])

$$i \partial_t \psi(x, t) = \left[-\frac{1}{2\mu} \partial_x^2 - \kappa \mu^2 \int dy U_\epsilon(x - y) |\psi(y, t)|^2 \right] \psi(x, t), \quad (2)$$

where κ characterizes the self-gravitational coupling, and $U_\epsilon(r)$ is a one-dimensional softened inverse-distance kernel,

$$U_\epsilon(r) = \frac{1}{\sqrt{r^2 + \epsilon^2}}, \quad (3)$$

with $\epsilon > 0$. This choice is a softened three-dimensional inverse-distance interaction restricted to a one-dimensional line (cf. [33]), avoiding the linear growth $|r|/2$ of the strict one-dimensional Poisson Green's function. All attractive gravitational terms are defined with explicit minus signs, with $U_\epsilon(r) > 0$. The strictly one-dimensional geometry enforces head-on encounters at zero impact parameter, which enhances interaction effects relative to generic three-dimensional scattering. While the quantitative peak values of the generated entanglement depend on the short-distance cutoff ϵ , the qualitative entanglement hierarchy we observe is primarily controlled by phase-space overlap and initial spatial geometry.

To generalize this to a bipartite system, we treat the two masses as distinguishable subsystems with coordinates x_1 and x_2 . We do not impose bosonic or fermionic exchange symmetry. We propose the following two-body equation of motion:

$$i \partial_t \Psi(x_1, x_2, t) = \left[-\frac{1}{2\mu_1} \partial_{x_1}^2 - \frac{1}{2\mu_2} \partial_{x_2}^2 - \kappa \mu_1 \Phi_1(x_1, t) - \kappa \mu_2 \Phi_2(x_2, t) - \gamma \mu_1 \mu_2 U_\epsilon(x_1 - x_2) \right] \Psi(x_1, x_2, t), \quad (4)$$

with local gravitational self-potentials

$$\Phi_1(x_1, t) = \int U_\epsilon(x_1 - x'_1) \mu_1 \rho_1(x'_1, t) dx'_1, \quad (5)$$

$$\Phi_2(x_2, t) = \int U_\epsilon(x_2 - x'_2) \mu_2 \rho_2(x'_2, t) dx'_2, \quad (6)$$

sourced dynamically by the marginal densities,

$$\rho_1(x_1, t) = \int |\Psi(x_1, x_2, t)|^2 dx_2, \quad \rho_2(x_2, t) = \int |\Psi(x_1, x_2, t)|^2 dx_1, \quad (7)$$

of the normalized two-particle wavefunction, $\int dx_1 dx_2 |\Psi(x_1, x_2, t)|^2 = 1$. The direct pair potential is a standard linear two-body operator:

$$V_{\text{pair}}(x_1, x_2) = -\gamma \mu_1 \mu_2 U_\epsilon(x_1 - x_2). \quad (8)$$

Here, the parameters κ and γ control the self-gravitational and pair-gravitational couplings, respectively. While physically there is only one Newton constant ($\kappa = \gamma$), we retain distinct symbols to distinguish nonlinear self-energy renormalization (κ) from linear inter-particle pair interaction (γ). We set $\hbar = 1$ and measure lengths, masses, and times in units x_0 , m_0 , and $t_0 = m_0 x_0^2 / \hbar$, so that the dimensionless masses are $\mu_i = m_i / m_0$ and the dimensionless couplings κ , γ absorb the physical Newton constant and the chosen regularization scale.

The phenomenological equation of motion (4) admits the following conserved total energy functional:

$$\begin{aligned} E[\Psi] = & \int dx_1 dx_2 \Psi^* (T_1 + T_2 + V_{\text{pair}}) \Psi \\ & - \frac{\kappa}{2} \mu_1^2 \iint \rho_1(x) U_\epsilon(x - x') \rho_1(x') dx dx' \\ & - \frac{\kappa}{2} \mu_2^2 \iint \rho_2(x) U_\epsilon(x - x') \rho_2(x') dx dx', \end{aligned} \quad (9)$$

where $T_i = -(2\mu_i)^{-1}\partial_{x_i}^2$ are the kinetic energy operators, and the factors of $1/2$ correctly prevent double-counting of the self-energy. Formula (9) shows that the effective generator of dynamics (4) is a time- and state-dependent ‘Hamiltonian’:

$$H_{\text{eff}}(t) = H_1[\rho_1](t) \otimes I_2 + I_1 \otimes H_2[\rho_2](t) + V_{\text{pair}}, \quad (10)$$

where the nonlinear single-particle Schrödinger–Newton operators are

$$H_i[\rho_i](t) = -\frac{1}{2\mu_i}\partial_{x_i}^2 - \kappa\mu_i\Phi_i(x_i, t). \quad (11)$$

The self-field terms Φ_i are nonlinear functionals of the state, but their operator structure is local. Setting $\gamma = 0$, the generator has the form

$$H_{\text{self}}(t) = A[\rho_1](t) \otimes I + I \otimes B[\rho_2](t),$$

with A and B Hermitian along the solution. The reduced density operator (whose diagonal elements yield the spatial density $\rho_1(x_1) = \rho_1^{\text{red}}(x_1, x_1)$) then obeys

$$\partial_t \rho_1^{\text{red}} = -i[A[\rho_1](t), \rho_1^{\text{red}}],$$

because the partial trace of the B commutator vanishes. Hence, for any integer $n \geq 1$,

$$\frac{d}{dt} \text{Tr}[(\rho_1^{\text{red}})^n] = -in \text{Tr}[(\rho_1^{\text{red}})^{n-1} [A[\rho_1](t), \rho_1^{\text{red}}]] = 0.$$

Thus the self-field flow strictly preserves the eigenvalue spectrum of the reduced density operator. It may change the Schmidt modes, but it cannot change the Schmidt eigenvalues.

The direct pair term V_{pair} is linear in Ψ but nonseparable as an operator on $\mathcal{H}_1 \otimes \mathcal{H}_2$. For the product initial states considered below, the residual nonadditive part of this operator has nonzero covariance, and the pair term therefore changes the Schmidt spectrum at order t^2 .

To quantify this entanglement generation, we first write the two-body state in its Schmidt decomposition:

$$\Psi(x_1, x_2, t) = \sum_k \sqrt{\lambda_k(t)} u_k(x_1) v_k(x_2), \quad (12)$$

where $u_k(x_1)$ and $v_k(x_2)$ are the orthonormal spatial Schmidt modes for each particle. The reduced density operator of particle 1 then has the kernel

$$\rho_1^{\text{red}}(x_1, x'_1; t) = \int \Psi(x_1, x_2, t) \Psi^*(x'_1, x_2, t) dx_2 = \sum_k \lambda_k(t) u_k(x_1) u_k^*(x'_1), \quad (13)$$

with real, non-negative Schmidt eigenvalues $\lambda_k(t)$ satisfying $\sum_k \lambda_k = 1$.

The von Neumann entanglement entropy reads

$$S_{\text{vN}}(t) = -\text{Tr}[\rho_1^{\text{red}}(t) \ln \rho_1^{\text{red}}(t)] = -\sum_k \lambda_k(t) \ln \lambda_k(t). \quad (14)$$

For a product initial state at $t = 0$, $\Psi(x_1, x_2, 0) = \psi_1(x_1, 0) \psi_2(x_2, 0)$, the reduced density operator is a rank-one projector $\rho_1^{\text{red}}(x_1, x'_1; 0) = \psi_1(x_1, 0) \psi_1^*(x'_1, 0)$, and therefore the initial entanglement is zero, $S_{\text{vN}}(0) = 0$.

For product initial states, separability-preserving mean-field projections identically yield $S_{\text{vN}}(t) = 0$. This effect was also observed for the canonical 2-particle Schrödinger–Newton equation [32] and is

analyzed in full generality in Appendix A. In contrast, the nonseparable pair interaction generically yields $S_{\text{vN}}(t) > 0$.

For smooth product initial states $\Psi(x_1, x_2, 0) = \psi_1(x_1, 0)\psi_2(x_2, 0)$, the linear entropy $S_L(t) = 1 - \text{Tr}[\rho_1^{\text{red}}(t)^2]$ vanishes at $t = 0$ together with its first derivative. The leading-order entanglement generation is therefore quadratic in time:

$$S_L(t) = 2t^2 [\mathbb{V}_{12}(V_{\text{pair}}) - \mathbb{V}_1(\langle V_{\text{pair}} \rangle_2) - \mathbb{V}_2(\langle V_{\text{pair}} \rangle_1)] + O(t^3), \quad (15)$$

where $\mathbb{V}_{12}$ denotes the variance in the joint product state, and $\mathbb{V}_1, \mathbb{V}_2$ denote variances over the respective marginals. Explicitly, these are defined as:

$$\mathbb{V}_{12}(V_{\text{pair}}) = \langle V_{\text{pair}}^2 \rangle_{12} - \langle V_{\text{pair}} \rangle_{12}^2, \quad (16)$$

$$\mathbb{V}_1(\langle V_{\text{pair}} \rangle_2) = \langle \langle V_{\text{pair}} \rangle_2^2 \rangle_1 - \langle \langle V_{\text{pair}} \rangle_2 \rangle_1^2, \quad (17)$$

$$\langle V_{\text{pair}} \rangle_2(x_1) = \int |\psi_2(x_2)|^2 V_{\text{pair}}(x_1, x_2) dx_2. \quad (18)$$

Here, $\langle V_{\text{pair}} \rangle_{1,2}$ denote partial expectations over the respective subsystems, with the symmetric definition

$$\langle V_{\text{pair}} \rangle_1(x_2) = \int |\psi_1(x_1)|^2 V_{\text{pair}}(x_1, x_2) dx_1. \quad (19)$$

Defining the nonadditive residual interaction

$$V_{\text{res}}(x_1, x_2) = V_{\text{pair}}(x_1, x_2) - \langle V_{\text{pair}} \rangle_2(x_1) - \langle V_{\text{pair}} \rangle_1(x_2) + \langle V_{\text{pair}} \rangle_{12}, \quad (20)$$

one may write

$$S_L(t) = 2t^2 \langle V_{\text{res}}^2 \rangle_{12} + O(t^3). \quad (21)$$

The coefficient is manifestly non-negative and vanishes when V_{pair} is additive on the support of the initial product state, $V_{\text{pair}}(x_1, x_2) = a(x_1) + b(x_2)$, since then $V_{\text{res}} = 0$ and purely local contributions to the interaction do not generate entanglement. We use the linear entropy S_L for this short-time expansion to avoid $t^2 \ln t$ non-analyticities associated with the von Neumann entropy.

2.1 Phase-space Wigner diagnostics

The entanglement diagnostics defined above are based on the Schmidt decomposition, which captures mode occupations but provides no spatial or momentum resolution. We use reduced Wigner functions to distinguish simple squeezing from wavepacket interference and spatial fragmentation.

For the reduced density operator $\hat{\rho}_1$ of particle 1 with position-space kernel $\rho_1^{\text{red}}(x, x')$ defined in Eq. (13), the Wigner function is

$$W(x, p) = \frac{1}{\pi} \int dy \rho_1^{\text{red}}(x + y, x - y) e^{-2ipy}, \quad (22)$$

where x and p are the phase-space coordinates of particle 1, and the integral runs over the displacement variable y . This definition is normalized so that $\int dp W(x, p) = \rho_1(x)$ and $\int dx dp W(x, p) = 1$. While pure Gaussian states possess strictly non-negative Wigner functions, mixed reduced states can exhibit Wigner negativity. Here we use $W(x, p)$ as a phase-space diagnostic of the local non-Gaussian structure associated with the Schmidt-spectrum evolution.

The single-particle reduced Wigner functions visualize the local phase-space state of each subsystem, but they are insensitive to correlations between the two particles. A more interaction-adapted diagnostic is obtained by tracing over the center-of-mass degree of freedom and Wigner-transforming the relative coordinate, since the pair potential depends only on $r = x_1 - x_2$.

We introduce center-of-mass and relative coordinates for equal masses ($\mu_1 = \mu_2$):

$$X_{\text{cm}} = \frac{1}{2}(x_1 + x_2), \quad r = x_1 - x_2, \quad (23)$$

with conjugate momenta $P = p_1 + p_2$ (CM) and $p_{\text{rel}} = \frac{1}{2}(p_1 - p_2)$ (relative). The full two-body Wigner function $W^{(2)}(X_{\text{cm}}, P, r, p_{\text{rel}})$ is a four-dimensional quasiprobability distribution. Marginalizing over the CM degree of freedom defines the *relative-coordinate reduced density operator*

$$\rho_{\text{rel}}(r, r') = \int dX_{\text{cm}} \Psi(X_{\text{cm}} + \frac{r}{2}, X_{\text{cm}} - \frac{r}{2}) \Psi^*(X_{\text{cm}} + \frac{r'}{2}, X_{\text{cm}} - \frac{r'}{2}), \quad (24)$$

and its Wigner transform

$$W_{\text{rel}}(r, p_{\text{rel}}) = \frac{1}{\pi} \int ds \rho_{\text{rel}}(r + s, r - s) e^{-2ip_{\text{rel}}s} \quad (25)$$

is the marginal of $W^{(2)}$ over the CM phase space: $W_{\text{rel}}(r, p_{\text{rel}}) = \int dX_{\text{cm}} dP W^{(2)}(X_{\text{cm}}, P, r, p_{\text{rel}})$. The normalization $\int dr dp_{\text{rel}} W_{\text{rel}} = 1$ holds, and the marginal $\int dp_{\text{rel}} W_{\text{rel}}(r, p_{\text{rel}}) = \rho_{\text{rel}}(r, r)$ gives the probability density of finding the two particles at relative separation r . Because V_{pair} acts only on r , $W_{\text{rel}}(r, p_{\text{rel}})$ is the most direct phase-space diagnostic of the degree of freedom driven by the entangling interaction. The generation of negativity in W_{rel} indicates non-Gaussian phase-space fragmentation, which accompanies, but does not by itself quantify, the growth of the Schmidt spectrum.

For unequal masses (Sec. 3.3), we use the mass-weighted center-of-mass coordinate $X_{\text{cm}} = (\mu_1 x_1 + \mu_2 x_2)/(\mu_1 + \mu_2)$ with conjugate momenta $P = p_1 + p_2$ and $p_{\text{rel}} = (\mu_2 p_1 - \mu_1 p_2)/(\mu_1 + \mu_2)$. The relative coordinate remains $r = x_1 - x_2$, and the inverse transformation is

$$x_1 = X_{\text{cm}} + \frac{\mu_2}{\mu_1 + \mu_2} r, \quad x_2 = X_{\text{cm}} - \frac{\mu_1}{\mu_1 + \mu_2} r. \quad (26)$$

For unequal masses, Eq. (24) is modified by using this inverse map inside the wavefunction arguments.

This separation will be used throughout the simulations below: κ changes the marginal density profiles entering the scattering problem, whereas γ supplies the nonseparable operator that changes the Schmidt spectrum. In the Appendix, we show that projecting the same pair-interaction model onto the Hartree product manifold recovers the separability-preserving coupled mean-field structure of Ref. [32]. Thus the absence of entanglement generation in Hartree mean-field equations follows from the product-state restriction rather than from the local SN self-field itself.

3 Numerical results

The evolution is computed using second-order Strang splitting on a one-dimensional periodic grid of length $L = 40$; the nonlinear self-potentials are recomputed from the instantaneous marginal densities at each potential substep. The gravitational potentials are evaluated via fast Fourier transform convolutions. The convolution entering the self-potentials is evaluated with the same periodic distance convention as the split-step grid; the pair kernel $U_{\epsilon}(x_i - x_j)$ is evaluated using the same periodic minimum-distance convention. The simulation domain is chosen large enough that the density remains negligible near the boundaries over the reported time window. Direct checks with $L = 60$ confirmed that periodic-image effects do not affect the quoted entropy values at the stated accuracy. We use $\epsilon = 0.2$, which provides a physically sharp pair-scattering cutoff while remaining robustly resolved on the spatial grid ($dx \approx 0.156$). The Schmidt spectrum is computed by performing a singular value decomposition (SVD) directly on the discretized two-particle wavefunction.

Unless stated otherwise, the baseline simulations use equal masses $\mu_1 = \mu_2 = 1$, initial mean separation $R_0 = |\langle x_1 \rangle - \langle x_2 \rangle|_{t=0} = 6$, width parameter $\sigma_0 = 1$, and a universal gravitational coupling $\lambda \equiv \kappa = \gamma = 1$. Quoted peak entropies refer to the plotted window $t \in [0, 40]$ unless a longer time horizon is explicitly indicated. The total energy $E_{\text{total}} = E_K + E_{\text{int}} + E_{\text{self}}$ is monitored throughout all runs. The relative energy drift satisfies $\max_t |E(t) - E(0)|/|E(0)| < 10^{-8}$, and the norm drift remained below 10^{-13} . Convergence was verified by doubling the grid ($N = 256 \rightarrow 512$), extending the domain ($L = 40 \rightarrow 60$), and halving the time step ($\Delta t = 0.01 \rightarrow 0.005$); the peak entropy values changed by less than 10^{-4} in absolute terms, and final separations shifted by less than 10^{-3} . The quoted entropy values are therefore insensitive to the grid, domain size, and time step at the precision reported here.

We use two interparticle separation diagnostics. For the localized-product collision, we report the marginal center-of-mass separation

$$\Delta x_{\text{mean}}(t) \equiv |\langle x_1 \rangle_t - \langle x_2 \rangle_t|. \quad (27)$$

For superposition states whose marginals can be symmetric by construction, we additionally report the rms relative separation

$$D_{\text{rel}}(t) = \sqrt{\langle (x_1 - x_2)^2 \rangle_t}, \quad (28)$$

which accurately measures the true interparticle distance even when the state is spatially delocalized over multiple distinct regions, a feature that the simple marginal difference Δx_{mean} cannot resolve. Here, D_{rel} represents the true quantum mechanical observable for the root-mean-square distance.

3.1 Initial state configurations

We prepare each initial wavepacket either as a Gaussian or as the stationary ground state of the single-particle SN equation.

The stationary single-particle SN profiles used as initial data are obtained by solving the nonlinear eigenproblem

$$\omega \phi_{\text{SN}}(x) = \left[-\frac{1}{2\mu} \partial_x^2 - \kappa \mu^2 \int U_\epsilon(x - x') |\phi_{\text{SN}}(x')|^2 dx' \right] \phi_{\text{SN}}(x), \quad (29)$$

subject to $\int |\phi_{\text{SN}}(x)|^2 dx = 1$. These states are obtained via imaginary-time propagation starting from a Gaussian seed of width σ_0 ; for the ground-state component used here, the converged profile is independent of the seed within numerical tolerance.

We construct the two-body initial state from these single-particle states in four distinct configurations. The localized single-particle components $\phi_L(x)$ and $\phi_R(x)$ are prepared by centering the chosen profile at $-R_0/2$ and $+R_0/2$, respectively. For Gaussian wavepackets we use

$$\phi_{L/R}(x) \propto \exp\left[-(x \pm R_0/2)^2/(4\sigma_0^2)\right]. \quad (30)$$

For stationary-profile simulations, we obtain the single-particle ground state $\phi_{\text{SN}}(x)$ by imaginary-time propagation and define the separated states by rigid translation:

$$\phi_{L/R}(x) = \phi_{\text{SN}}(x \pm R_0/2). \quad (31)$$

Their initial spatial overlap is $s \equiv \int dx \phi_L^*(x) \phi_R(x)$, which for Gaussians is $s = \exp[-R_0^2/8\sigma_0^2]$. For the real translated profiles used here, $s > 0$.

Using these well-separated left (L) and right (R) spatial modes, we investigate the following four initial two-body configurations:

$$\Psi_{\text{I}}(x_1, x_2) = \phi_L(x_1)\phi_R(x_2), \quad (32)$$

$$\Psi_{\text{II}}(x_1, x_2) = \mathcal{N}_{\text{II}}[\phi_L(x_1) + \phi_R(x_1)][\phi_L(x_2) + \phi_R(x_2)], \quad (33)$$

$$\Psi_{\text{III}}(x_1, x_2) = \mathcal{N}_{\text{III}}[\phi_L(x_1)\phi_L(x_2) + \phi_R(x_1)\phi_R(x_2)], \quad (34)$$

$$\Psi_{\text{IV}}(x_1, x_2) = \mathcal{N}_{\text{IV}}[\phi_L(x_1)\phi_R(x_2) + \phi_R(x_1)\phi_L(x_2)], \quad (35)$$

where the normalizations are $\mathcal{N}_{\text{II}} = [4(1+s)^2]^{-1/2}$ and $\mathcal{N}_{\text{III}} = \mathcal{N}_{\text{IV}} = [2+2s^2]^{-1/2}$.

The localized product Ψ_{I} is purely separable ($S_{\text{vN}} = 0$). The delocalized product Ψ_{II} is also separable, but for negligible overlap s , the marginal densities are split equally between the two wells. The superpositions Ψ_{III} and Ψ_{IV} are Bell-like entangled states; their two nonzero Schmidt eigenvalues are $\lambda_{\pm} = (1 \pm s)^2/[2(1+s^2)]$, giving $S_{\text{vN}}(0) = -\lambda_+ \ln \lambda_+ - \lambda_- \ln \lambda_- \rightarrow \ln 2$ in the negligible-overlap limit $s \rightarrow 0$. Ψ_{III} features position-correlated particles, while in Ψ_{IV} the particles are anti-correlated.

3.2 Equal-mass collision dynamics

In the following, we analyze the collision dynamics for equal masses ($\mu_1 = \mu_2 = 1$) across the four initial configurations.

3.2.1 Localized product

The localized product state Ψ_{I} (Eq. 32) has $S_{\text{vN}}(0) = 0$ and a factorized joint probability density $|\Psi_{\text{I}}|^2 = \rho_1(x_1)\rho_2(x_2)$ representing two wavepackets centered at opposite sides.

The two wavepackets accelerate toward each other under mutual gravitational attraction and collide. The collision dynamics depend on the single-particle profile (Figs. 1 and 2).

Gaussian wavepackets are not stationary states of the self-gravitational potential and undergo kinetic dispersion. The broadened distributions overlap extensively, generating entanglement ($S_{\text{vN}} \approx 0.87$). After the collision, the Gaussians merge at the center of mass ($\Delta x_{\text{mean}} \rightarrow 0$), coalescing into a single central peak with the entropy saturating at its post-collision value. The merger is effectively inelastic at the level of the reduced one-body density: relative translational kinetic energy is transferred into internal dispersive degrees of freedom, while total energy and norm remain conserved.

Stationary SN profiles behave differently. Their self-bound structure is perturbed only weakly during the collision. The particles remain gravitationally bound, oscillating periodically in their mutual well. The entanglement per bounce is small ($S_{\text{vN}} \approx 0.19$ after a few collisions) but accumulates through repeated elastic scattering. Extended simulations to $t = 200$ confirm that the oscillation persists, with entanglement saturating at $S_{\text{vN}} \approx 0.29$.

At $t = 0$, the relative-coordinate Wigner function W_{rel} (bottom rows of Figs. 1 and 2) is a single localized lobe centered at $r = -R_0$. For stationary profiles, it remains predominantly positive throughout the collision, developing only mild phase-space shear. For dispersive Gaussians, the broadening prior to collision produces pronounced phase-space shearing and Wigner fragmentation.

3.2.2 Delocalized product

In the delocalized product configuration Ψ_{II} (Eq. 33), each particle is prepared in the same coherent superposition of left and right wave packets. Although Ψ_{II} has $S_{\text{vN}}(0) = 0$, its joint density has four equal-weight lobes corresponding to LL, LR, RL, and RR components (Fig. 3).

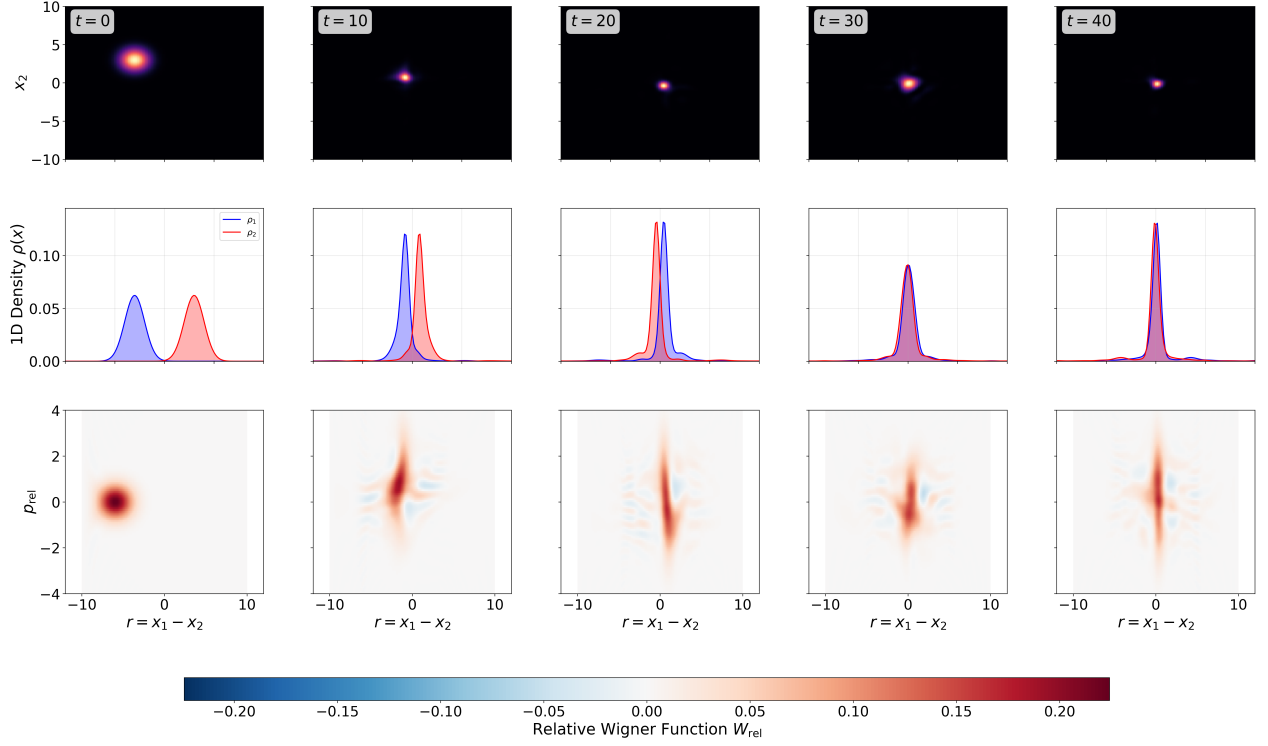


Figure 1: Collision dynamics for the localized product state Ψ_I (Eq. 32) using Gaussian wavepackets. The snapshots are taken at $t = 0, 10, 20, 30, 40$. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ (blue) and $\rho_2(x)$ (red); (bottom row) the relative-coordinate Wigner function W_{rel} . Parameters: $R_0 = 6$, $\kappa = \gamma = 1$, $\sigma_0 = 1$.

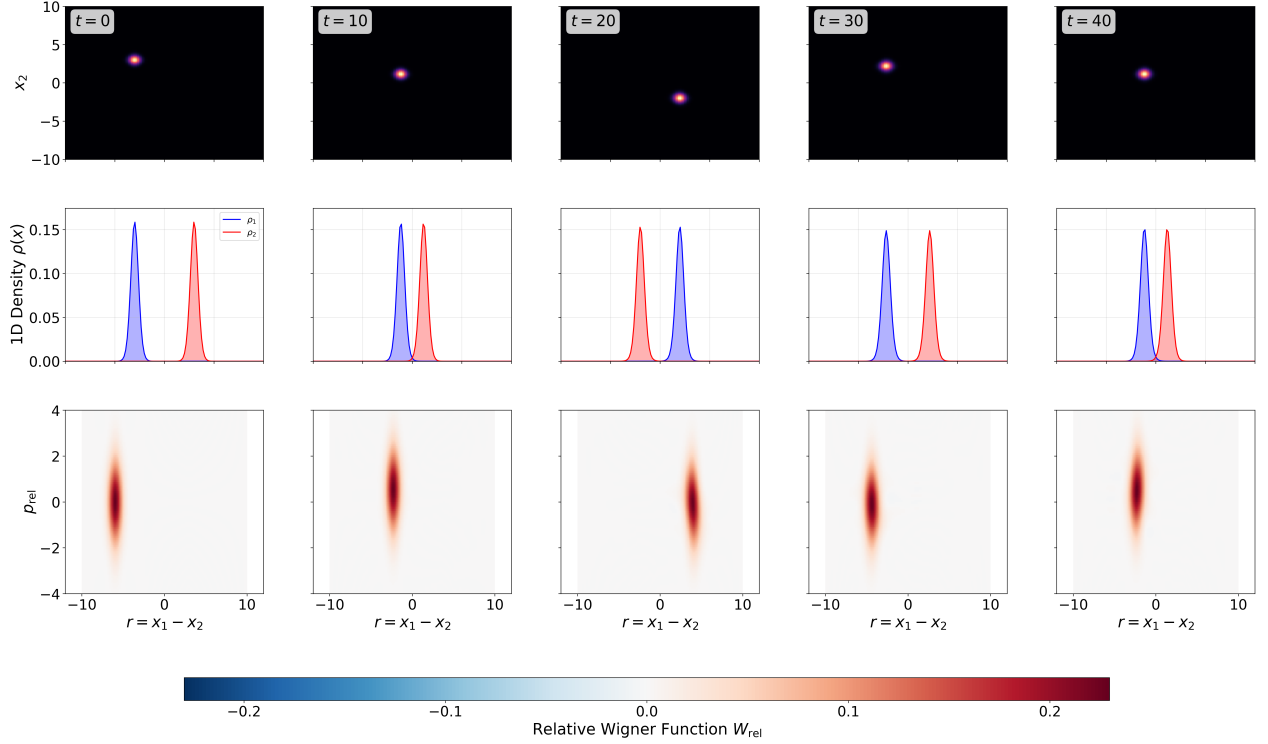


Figure 2: Collision dynamics for the localized product state Ψ_I (Eq. 32) using stationary SN profiles. The snapshots are taken at $t = 0, 10, 20, 30, 40$. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ (blue) and $\rho_2(x)$ (red); (bottom row) the relative-coordinate Wigner function W_{rel} . Parameters: $R_0 = 6$, $\kappa = \gamma = 1$; the imaginary-time seed width is $\sigma_0 = 1$.

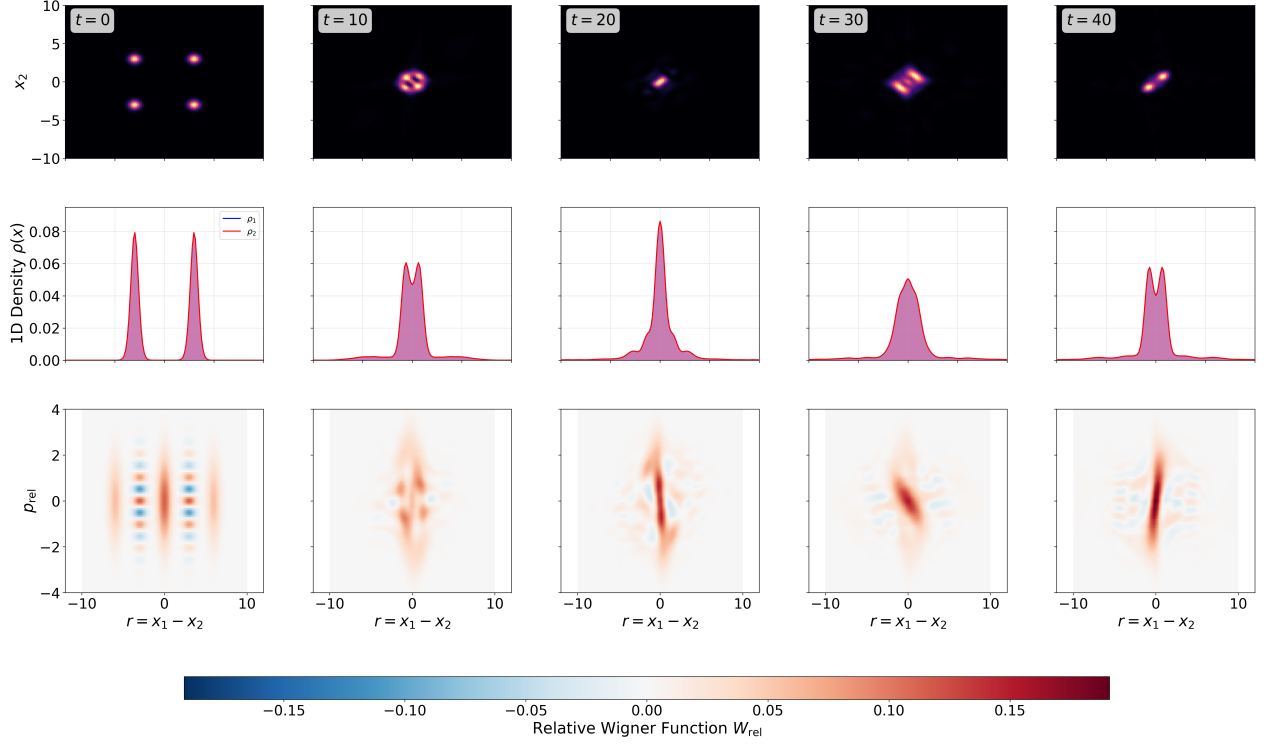


Figure 3: Evolution of the delocalized product state Ψ_{II} (Eq. 33) using stationary SN profiles. Despite being initialized from stationary profiles, the LL/RR and LR/RL components acquire different phases and accelerations under the pair potential, producing $S_{\text{vN}} \approx 1.5$. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ and $\rho_2(x)$; (bottom row) the relative-coordinate Wigner function W_{rel} , with snapshots at $t = 0, 10, 20, 30, 40$. Parameters: $R_0 = 6$, $\kappa = \gamma = 1$; the imaginary-time seed width is $\sigma_0 = 1$.

For negligible overlap s , the marginal density is split approximately equally between L and R , so the local part of the self-gravitational well at each component is sourced by only half the total density, weakening the local binding. The remote component generates a secondary attraction toward the center. The delocalized product reaches $S_{\text{vN}} \approx 1.58$, roughly an order of magnitude above the localized-product collision ($S_{\text{vN}} \approx 0.19$).

Although the local state for each particle initially spans a two-dimensional subspace $\{\phi_L, \phi_R\}$, the observed entanglement $S_{\text{vN}} \approx 1.58$ exceeds the rigid-profile maximum of $\ln 2 \approx 0.69$. This confirms that the full pair-interaction dynamics excites higher-order spatial modes well beyond the initial two-mode subspace.

The pair potential acts differently on the co-localized components ($\phi_L \otimes \phi_L$ and $\phi_R \otimes \phi_R$) and the separated components ($\phi_L \otimes \phi_R$ and $\phi_R \otimes \phi_L$), because the former sample the smallest separations, where $U_\epsilon(x_1 - x_2)$ is largest and the attractive pair potential is most negative.

In phase space, the reduced Wigner function $W(x_1, p_1)$ initially describes a pure coherent superposition $\psi_1(x) \propto \phi_L(x) + \phi_R(x)$, containing two localized positive lobes centered at $(\pm R_0/2, 0)$ and interference fringes between them. As the system evolves, the superposition-induced nonstationarity sets in: because the marginal density is split, each lobe experiences only half the self-gravitational binding depth. Simultaneously, the density of the remote spatial component generates a secondary attractive potential toward the center. The lobes accelerate inward, acquiring nonzero mean momenta $\langle p \rangle \neq 0$, visible as a tilt of the Wigner lobes away from the $p = 0$ axis. During the collision, the two counter-propagating phase-space distributions overlap and interfere. Small-scale interference fringes appear between the counter-propagating components, and the generation of extended negative regions $\mathcal{V}_- = \int dx dp |\min(W, 0)|$ accompanies the growth of the entanglement entropy.

In the relative coordinate (bottom row of Fig. 3), the initial state is a macroscopic superposition of co-localized ($r = 0$) and separated ($r = \pm R_0$) components, producing high-frequency interference fringes. The gravitational pair potential pulls the separated components toward the center, shearing the relative phase space and merging the components into a tilted central stripe. The resulting Wigner negativity diagnoses non-Gaussian phase-space fragmentation.

3.2.3 Correlated superposition

The correlated (“cat”) state Ψ_{III} (Eq. 34) is a coherent same-side superposition. Unlike the product states, this configuration is entangled at $t = 0$, with both particles localized on the same side (L, L or R, R), but the superposition cannot be factorized into single-particle states.

The gravitational self-interaction acts symmetrically on the two components (Fig. 4), and the entanglement entropy oscillates around $S_{\text{vN}} \approx 1.28$ without systematic growth or decay. The marginal center-of-mass separation $|\langle x_1 \rangle - \langle x_2 \rangle|$ remains near zero throughout the evolution by symmetry. However, examining the local density reveals non-stationary dynamics driven by the non-linearity of the SN equation. Because the self-potential $\Phi(x)$ is sourced by the marginal density, which is split equally between L and R (for negligible overlap s), the local part of the self-gravitational well at each branch is approximately half as deep as for an isolated stationary profile. The original profile is therefore no longer a stationary solution of this weakened potential and undergoes immediate kinetic dispersion. Simultaneously, the density of the remote spatial component generates a secondary gravitational attraction toward the center, accelerating the components into a collision at the origin. The entanglement rises from its initial value (close to $\ln 2$ for the numerically small overlap used here) to oscillate around $S_{\text{vN}} \approx 1.28$. Because this value exceeds the rigid-profile maximum of $\ln 2 \approx 0.69$, the full pair-interaction dynamics clearly excites higher-order spatial modes beyond the initial left/right subspace.

In phase space, both particles are always co-localized ($r \approx 0$), so the relative Wigner function

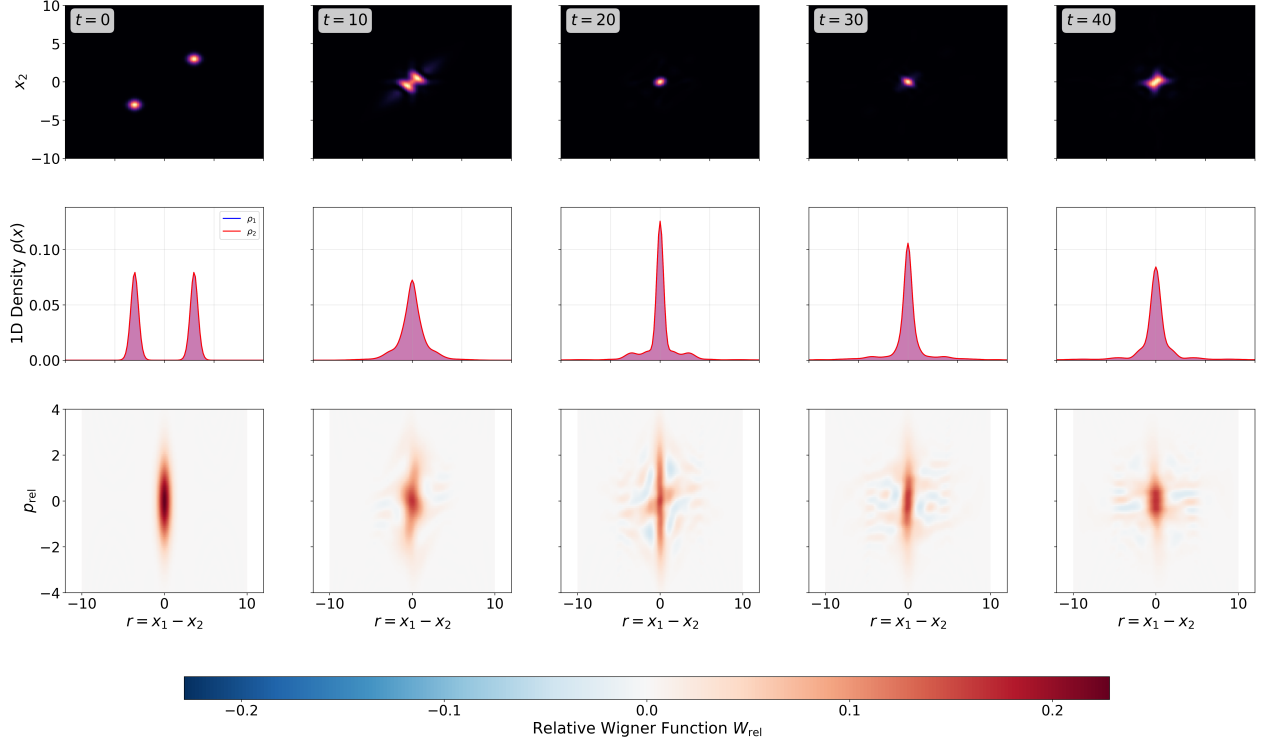


Figure 4: Evolution of the correlated superposition state Ψ_{III} (Eq. 34) using stationary SN profiles. The entanglement entropy oscillates around $S_{\text{vN}} \approx 1.28$, reflecting deformation of the two same-side components under the common self-field and pair attraction. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ and $\rho_2(x)$; (bottom row) the relative-coordinate Wigner function W_{rel} , with snapshots at $t = 0, 10, 20, 30, 40$. Parameters: $R_0 = 6$, $\kappa = \gamma = 1$; the imaginary-time seed width is $\sigma_0 = 1$.

starts as a single localized lobe at the origin with no macroscopic interference fringes. Because the relative coordinate is already centered, the mutual gravitational force does not drive macroscopic transport in r . The state undergoes local squeezing as the marginals collapse, which explains why the entanglement oscillates at a moderate level without systematic growth.

3.2.4 Anticorrelated superposition

The anticorrelated configuration Ψ_{IV} (Eq. 35) places the particles on opposite sides. This state is also entangled at $t = 0$, but unlike the correlated case, its two components (L, R and R, L) are spatially disjoint.

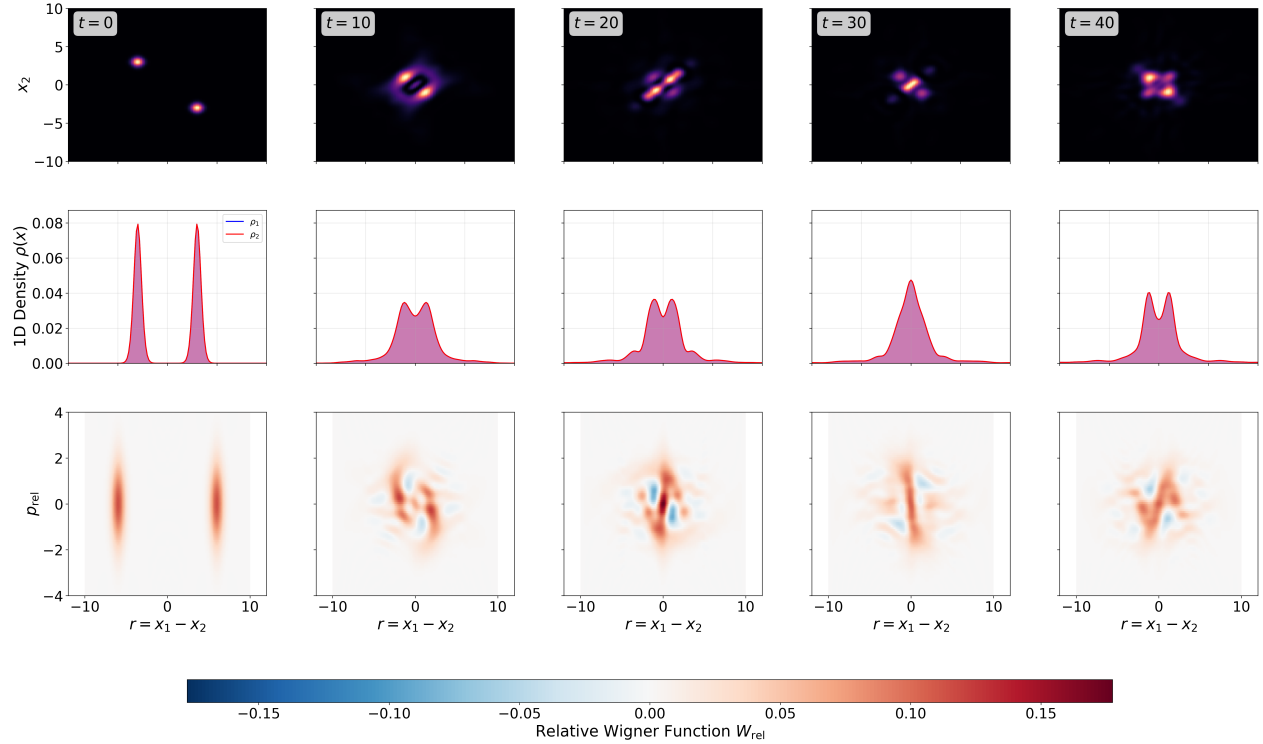


Figure 5: Evolution of the anticorrelated superposition state Ψ_{IV} (Eq. 35) using stationary SN profiles. The mutual gravitational attraction drives the disjoint spatial components together, increasing the entanglement entropy to $S_{\text{vN}} \approx 1.67$. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ and $\rho_2(x)$; (bottom row) the relative-coordinate Wigner function W_{rel} , with snapshots at $t = 0, 10, 20, 30, 40$. Parameters: $R_0 = 6$, $\kappa = \gamma = 1$; the imaginary-time seed width is $\sigma_0 = 1$.

In the anticorrelated superposition (Fig. 5), the mutual gravitational attraction actively pulls the particles in each component toward each other, driving the $r = \pm R_0$ relative-coordinate components toward overlap near $r = 0$. The entanglement entropy reaches $S_{\text{vN}} \approx 1.67$, comparable to the delocalized product, and the rms relative separation $D_{\text{rel}}(t)$ undergoes large-amplitude oscillations as the components interfere constructively and destructively during their gravitational evolution.

The relative coordinate is in a macroscopic superposition of $r = \pm R_0$. At $t = 0$, the relative Wigner function has two localized lobes at $r = \pm R_0$ separated by high-frequency interference fringes. The mutual gravitational force accelerates these components toward each other, shearing them in phase space. Their collision at the origin produces Wigner negativity.

3.2.5 Entanglement dynamics

Figure 6 compares $S_{\text{vN}}(t)$ and the leading Schmidt eigenvalues for Gaussian and stationary-profile initial data. Because the self-field is isospectral, all spectrum changes are due to V_{pair} .

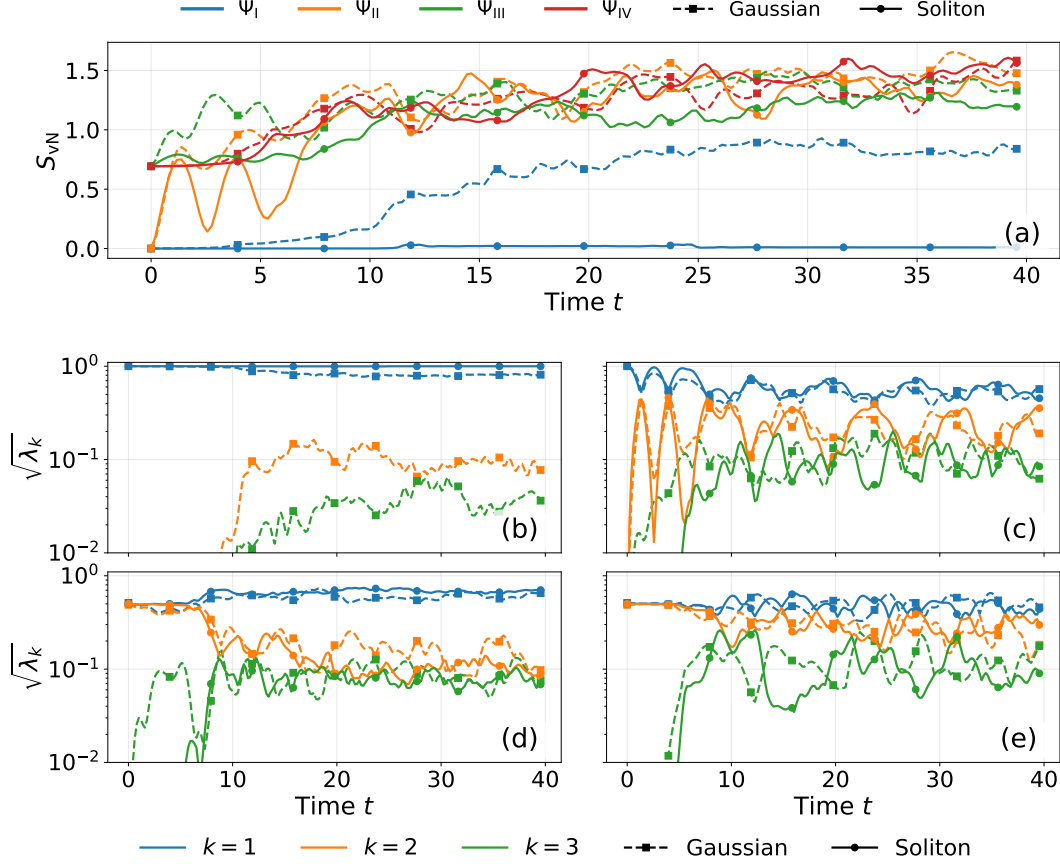


Figure 6: Entanglement and Schmidt spectrum dynamics comparing Gaussian wavepackets (solid lines) and stationary SN profiles (dashed lines with circular markers) across the four initial condition configurations. (a) Evolution of the von Neumann entanglement entropy S_{vN} . (b-e) Evolution of the Schmidt spectrum (eigenvalues λ_k) for the lowest three modes: (b) Localized product Ψ_I (Eq. (32)), (c) Delocalized product Ψ_{II} (Eq. (33)), (d) Correlated superposition Ψ_{III} (Eq. (34)), and (e) Anticorrelated superposition Ψ_{IV} (Eq. (35)). In all cases, stationary profiles suppress the transfer of weight into higher Schmidt modes compared with Gaussian packets, where kinetic spreading and collision-induced distortion accelerate Schmidt-spectrum broadening.

Among stationary-profile simulations, the localized product produces the smallest dynamically generated entropy, reaching $S_{\text{vN}} \approx 0.19$ over the plotted window. The correlated cat state starts close to $\ln 2$ and oscillates around $S_{\text{vN}} \approx 1.2$, while the anticorrelated and delocalized product states reach the largest absolute entropies ($S_{\text{vN}} \approx 1.58$ and 1.67), driven by the different pair-potential phases and accelerations sampled by co-localized and separated components. For the parameters studied here, the entropy hierarchy is controlled mainly by spatial geometry: states with multiple relative-coordinate components populate higher Schmidt modes more efficiently than localized product collisions.

Figure 6(b-e) shows the corresponding redistribution of Schmidt weight. The decay of λ_1 and

the growth of λ_2, λ_3 indicate occupation of spatial modes beyond the initial left/right subspace. Gaussian wavepackets show stronger redistribution because kinetic spreading increases the spatial support over which the pair potential varies. For stationary profiles, self-localization keeps the packets narrower and suppresses this transfer to higher Schmidt modes.

3.3 Mass asymmetry

We next consider unequal masses, keeping $\mu_2 = 1$ fixed and increasing μ_1 . We consider a localized-product collision with a mass ratio of 4:1 ($\mu_1 = 4.0$, $\mu_2 = 1.0$), using the universal coupling $\kappa = \gamma = 1.0$.

Figure 7 shows the joint density, marginals, and relative Wigner function for this mass-asymmetric collision. Owing to the steep μ^2 scaling of the self-gravitational well depth, the heavy stationary profile becomes narrow on the scale of the light profile and acts approximately as a localized scattering potential. By contrast, the lighter particle has a broader stationary profile with weaker self-binding. As the mutual attraction drives the packets together, the broad light profile scatters from this narrow heavy-mass potential.

Because the light particle's self-binding ($\kappa\mu_2^2 = 1.0$) is weaker than that of the heavy particle, the light stationary profile is deformed during the collision. The heavy profile acts as a sharp scattering center, exciting oscillatory distortions of the lighter marginal density. This deformation is visible in the relative Wigner function (bottom row of Fig. 7), which develops interference fringes near the collision. The corresponding entropy increase reaches $S_{\text{vN}} \approx 1.94$, indicating that large-scale fragmentation at mass ratio 4:1 produces substantial entanglement.

To quantify the broadening of the lighter marginal density, we use the Participation Ratio

$$\text{PR} = \left[\int \rho^2(x) dx \right]^{-1}, \quad (36)$$

which scales linearly with the spatial extent of the wavepacket. Unlike the variance, the PR weights the high-density support of the marginal and is less sensitive to small tails at large $|x|$.

Figure 8 (Left) maps the PR of the lighter particle's marginal density as a function of time and mass ratio. Increases in PR indicate spatial broadening and deformation of the light particle. For stationary SN profiles this broadening remains comparatively limited over the scanned range, whereas Gaussian packets display a much more persistent PR increase because kinetic dispersion exposes them to larger regions of the scattering potential.

For stationary SN profiles, self-binding limits the PR growth and the associated Schmidt-spectrum broadening; for Gaussian packets, prior kinetic spreading increases the spatial region over which the tidal gradient acts.

Figure 8 (Right) shows the peak entanglement $\max_t S_{\text{vN}}$ as a function of the mass ratio μ_1/μ_2 , with $\mu_2 = 1.0$ held fixed, for both Gaussian wavepackets and stationary SN profiles.

For stationary SN profiles, the peak entanglement remains nearly constant across the scanned mass ratios: self-binding suppresses spatial fragmentation during the collision. The freely dispersing Gaussian wavepackets, by contrast, undergo increasingly pronounced spatial disruption by the heavy mass, and the entanglement entropy grows steadily with μ_1/μ_2 .

4 Conclusions and Outlook

The question of entanglement generation through gravity is of primary interest in foundational studies [31]. It has been claimed that classical gravity cannot generate entanglement between spacelike separated particles and, consequently, that the observation of gravitationally-induced entanglement

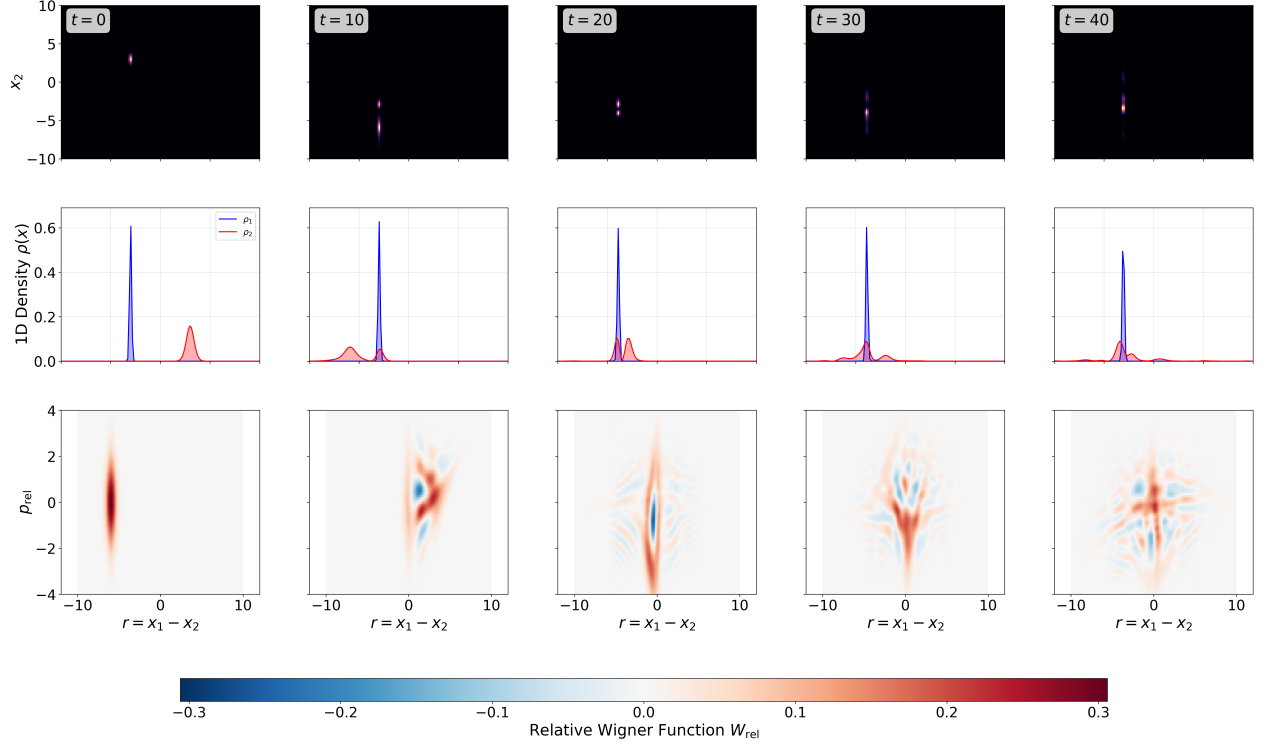


Figure 7: Mass-asymmetric localized-product collision with $\mu_1 = 4.0$ and $\mu_2 = 1.0$. The heavy particle (blue) remains nearly stationary while the light particle (red) is accelerated and deformed during the collision. The panel shows (top row) the joint probability density $|\Psi(x_1, x_2)|^2$; (middle row) single-particle marginal densities $\rho_1(x)$ and $\rho_2(x)$; (bottom row) the relative-coordinate Wigner function W_{rel} , with snapshots at $t = 0, 10, 20, 30, 40$. Parameters: $R_0 = 6$, $\kappa = \gamma = 1$; the imaginary-time seed width is $\sigma_0 = 1$.

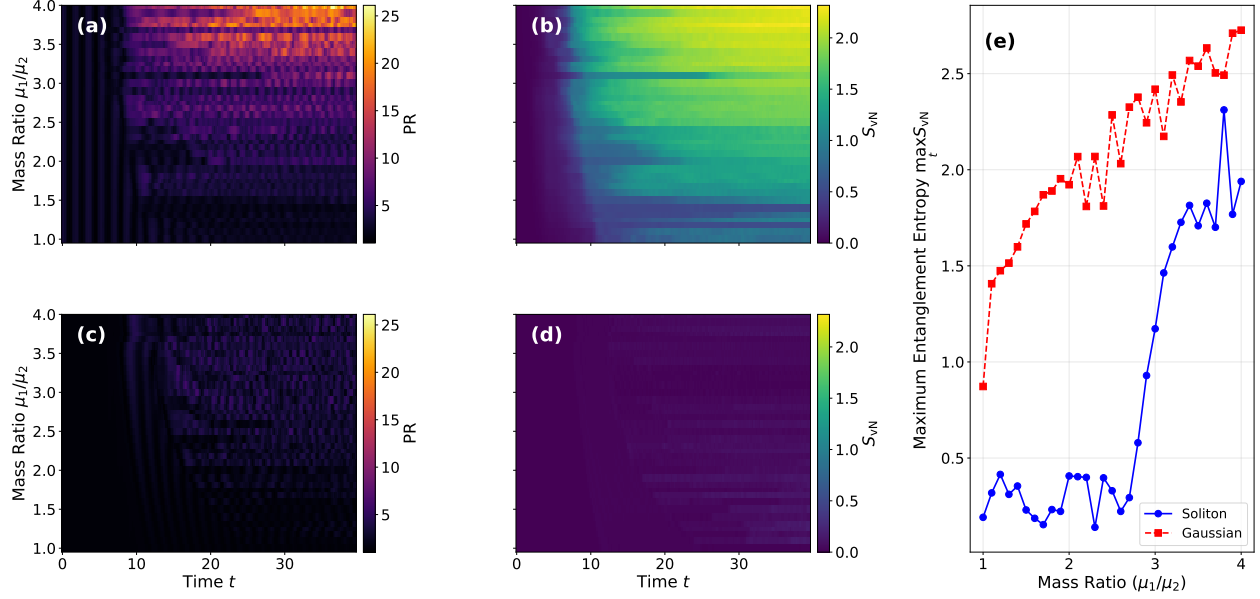


Figure 8: Gravitational shattering and entanglement generation in mass-asymmetric collisions. (a,c) Heatmaps of the Participation Ratio [PR, Eq. (36)] of the lighter particle's marginal density $\rho_2(x, t)$ as a function of time and mass ratio μ_1/μ_2 for Gaussian wavepackets (a) and stationary SN profiles (c). Spatial deformation appears as an increase in PR. The dispersing Gaussian packet (a) undergoes significant shattering, whereas the stationary profile (c) suppresses deformation. (b,d) Corresponding heatmaps of the von Neumann entanglement entropy $S_{vN}(t)$ for Gaussian (b) and stationary (d) initial states. Entanglement generation strongly correlates with the spatial disruption of the lighter mass. (e) Peak entanglement $\max_t S_{vN}$ (over $t \in [0, 40]$) versus mass ratio μ_1/μ_2 . Stationary-profile entanglement remains minimal and nearly constant across mass ratios, while Gaussian entanglement grows rapidly as the scattering potential acts on the dispersing, fragmented wavepacket.

would provide a decisive argument for the quantumness of the gravitational field (see [30, 34, 35, 36]). However, this argument hinges upon the assumption that the gravitational interaction is *local*. This can be questioned, given the fact there exist no local expression for the energy of the gravitational field [37]. While the single-particle SN equation is not local in the spacetime sense [35, 12], its canonical 2-particle version [1] *is* local in the sense of bipartite dynamics [32]. In Appendix A we show that this is generally the case in models admitting the Hartree projection onto the product-state manifold.

We analyzed a two-body Schrödinger–Newton-type model that explicitly separates gravitational entanglement generation from mean-field self-localization. The numerical analysis demonstrates that entanglement growth depends sensitively on spatial dispersion. Highly localized, self-bound profiles undergo quasi-elastic scattering with minimal Schmidt-spectrum excitation. Kinetically dispersing wavepackets amplify the entangling power of the pair potential. In asymmetric collisions, the gravitational field of the heavier mass spatially disrupts the lighter wavepacket, producing pronounced Wigner negativity and rapid entanglement growth. Consequently, any semiclassical formulation combining Diósi–Penrose self-localization with Newtonian mutual attraction must explicitly define whether the interaction preserves separability as a mean field or operates as a nonseparable pair interaction.

The most direct extensions are 2D and 3D scattering geometries with finite impact parameter, N -body collisions, and the addition of an external disorder potential to study the interplay with Anderson-type localization. Furthermore, developing realistic physical models will require moving beyond these simplified low-dimensional head-on collisions to full 3D simulations. Such extensions are crucial for determining whether this direct gravitational entanglement generation could ever be isolated and measured in next-generation optomechanical levitation experiments.

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A Hartree reduction and recovery of mean-field models

The separability-preserving Schrödinger–Newton formulation studied in Ref. [32] is characterized by gravitational interactions that preserve the product form of initially product many-body states. Within the present variational hierarchy, the same separability-preserving structure is obtained as the Hartree projection of the pair-interaction dynamics onto the product-state manifold via the Dirac–Frenkel variational principle [38, 39].

We impose the Hartree product ansatz on the pair-interaction model:

$$\Psi(x_1, x_2, t) = \psi_1(x_1, t)\psi_2(x_2, t), \quad (37)$$

with marginals $\rho_i(x_i, t) = |\psi_i(x_i, t)|^2$ normalized to unity. To derive the mean-field limit, we constrain the system to the Hartree manifold—the separable submanifold of $\mathcal{H}_1 \otimes \mathcal{H}_2$ spanned by exact product states. Rather than linearizing the SN self-field, we restrict the full energy functional associated with Eq. (4) to this zero-entanglement geometry. For a factorized state $\Psi = \psi_1\psi_2$, the functional decomposes as

$$E[\psi_1, \psi_2] = E_1[\psi_1] + E_2[\psi_2] + E_{\text{pair}}[\psi_1, \psi_2], \quad (38)$$

Because the state is an exact product, the kinetic and self-energy terms decouple trivially, yielding local energy terms for each particle:

$$E_1[\psi_1] = \int dx_1 \psi_1^*(x_1, t) \left(-\frac{1}{2\mu_1} \partial_{x_1}^2 \right) \psi_1(x_1, t) - \frac{\kappa}{2} \mu_1^2 \iint dx_1 dx'_1 |\psi_1(x_1, t)|^2 U_\epsilon(x_1 - x'_1) |\psi_1(x'_1, t)|^2, \quad (39)$$

and similarly for $E_2[\psi_2]$. The factor of 1/2 prevents double-counting of the self-energy. The expectation value of the nonseparable pair interaction V_{pair} evaluated on this product manifold reduces to the density-density interaction energy between the two marginal densities:

$$E_{\text{pair}}[\psi_1, \psi_2] = -\gamma \mu_1 \mu_2 \int dx_1 dx_2 |\psi_1(x_1, t)|^2 |\psi_2(x_2, t)|^2 U_\epsilon(x_1 - x_2). \quad (40)$$

To obtain the equations of motion for the individual factors ψ_1 and ψ_2 , we apply the Dirac–Frenkel–McLachlan time-dependent variational principle [38, 39, 40] to the action

$$\mathcal{S}[\Psi] = \int dt \left[i \int dx_1 dx_2 \Psi^* \partial_t \Psi - E[\Psi] \right], \quad (41)$$

where $E[\Psi]$ is the conserved energy functional. This formulation is essential for the nonlinear SN terms: using the expectation value of the instantaneous effective Hamiltonian would double-count the self-gravitational energy. For the restricted product-state manifold, we demand stationarity under independent variations $\delta\psi_1^*$ and $\delta\psi_2^*$.

For a normalized product state $\Psi = \psi_1\psi_2$, the expectation value of the time-derivative operator expands as:

$$\int dx_1 dx_2 \Psi^* i \partial_t \Psi = i \int dx_1 \psi_1^* \partial_t \psi_1 + i \int dx_2 \psi_2^* \partial_t \psi_2. \quad (42)$$

The variations are taken tangent to the normalized Hartree manifold; equivalently, one may introduce time-dependent Lagrange multipliers enforcing $\|\psi_1\| = \|\psi_2\| = 1$, which only modify the gauge terms below. Setting the variational derivative of the action with respect to $\psi_1^*(x_1, t)$ to zero yields:

$$i \partial_t \psi_1(x_1, t) + i \left(\int dx_2 \psi_2^* \partial_t \psi_2 \right) \psi_1(x_1, t) = \frac{\delta}{\delta \psi_1^*(x_1, t)} E[\psi_1, \psi_2]. \quad (43)$$

We evaluate the functional derivatives $\delta E/\delta\psi_1^*$ term by term. For the single-particle functional E_1 , the variation with respect to ψ_1^* yields the standard nonlinear Schrödinger–Newton term, where the factor of 1/2 is cancelled by the product rule applied to the quartic interaction $|\psi_1|^2|\psi_1'|^2$:

$$\begin{aligned}\frac{\delta E_1}{\delta\psi_1^*(x_1, t)} &= \left[-\frac{1}{2\mu_1}\partial_{x_1}^2 - \kappa\mu_1^2 \int U_\epsilon(x_1 - x'_1)|\psi_1(x'_1, t)|^2 dx'_1 \right] \psi_1(x_1, t) \\ &= \left[-\frac{1}{2\mu_1}\partial_{x_1}^2 - \kappa\mu_1\Phi_1(x_1, t) \right] \psi_1(x_1, t).\end{aligned}\quad (44)$$

Next, taking the variational derivative of the pair energy E_{pair} with respect to ψ_1^* yields an effective one-body potential generated by the partner’s density:

$$\frac{\delta E_{\text{pair}}}{\delta\psi_1^*(x_1, t)} = \left[-\gamma\mu_1\mu_2 \int U_\epsilon(x_1 - x_2)|\psi_2(x_2, t)|^2 dx_2 \right] \psi_1(x_1, t). \quad (45)$$

We identify the integral bracket as the effective mean-field cross-coupling:

$$\tilde{\Phi}_{12}^{(2\rightarrow 1)}(x_1, t) = \int U_\epsilon(x_1 - x_2)\rho_2(x_2, t) dx_2. \quad (46)$$

By symmetry, the corresponding variations with respect to ψ_2^* yield the analogous terms for the second particle, including the reciprocal mean-field potential:

$$\tilde{\Phi}_{12}^{(1\rightarrow 2)}(x_2, t) = \int U_\epsilon(x_1 - x_2)\rho_1(x_1, t) dx_1. \quad (47)$$

Combining these variations, the resulting Hartree-projected equations of motion are:

$$i\partial_t\psi_1(x_1, t) = \left[-\frac{1}{2\mu_1}\partial_{x_1}^2 - \kappa\mu_1\Phi_1(x_1, t) - \gamma\mu_1\mu_2\tilde{\Phi}_{12}^{(2\rightarrow 1)}(x_1, t) \right] \psi_1(x_1, t) + C_1(t)\psi_1(x_1, t), \quad (48)$$

$$i\partial_t\psi_2(x_2, t) = \left[-\frac{1}{2\mu_2}\partial_{x_2}^2 - \kappa\mu_2\Phi_2(x_2, t) - \gamma\mu_1\mu_2\tilde{\Phi}_{12}^{(1\rightarrow 2)}(x_2, t) \right] \psi_2(x_2, t) + C_2(t)\psi_2(x_2, t), \quad (49)$$

where the purely time-dependent phase constants are $C_1(t) = -i(\int dx_2 \psi_2^* \partial_t \psi_2)$ and $C_2(t) = -i(\int dx_1 \psi_1^* \partial_t \psi_1)$. These are pure gauge terms that can be set to zero by imposing the parallel-transport conditions $\int dx_i \psi_i^* \partial_t \psi_i = 0$.

Equations (48)–(49) have the same separability-preserving Hartree structure as the Schrödinger–Newton equations of Ref. [32]:

$$i\partial_t\psi_1 = \left[T_1 + V_{11}[|\psi_1|^2] + V_{12}[|\psi_2|^2] \right] \psi_1, \quad (50)$$

$$i\partial_t\psi_2 = \left[T_2 + V_{22}[|\psi_2|^2] + V_{21}[|\psi_1|^2] \right] \psi_2, \quad (51)$$

where V_{ii} are self-potentials and V_{ij} are mean-field cross-potentials. Thus, separability-preserving SN models arise naturally as the Hartree projection of the pair-interaction dynamics. This product reduction precludes entanglement generation by construction. For a universal coupling $\kappa = \gamma \equiv \lambda$, these projected equations match the product-state restriction of a semiclassical mean-field model where gravity is sourced by the expectation value $\langle \hat{\rho} \rangle$, up to the chosen dimensional reduction.

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